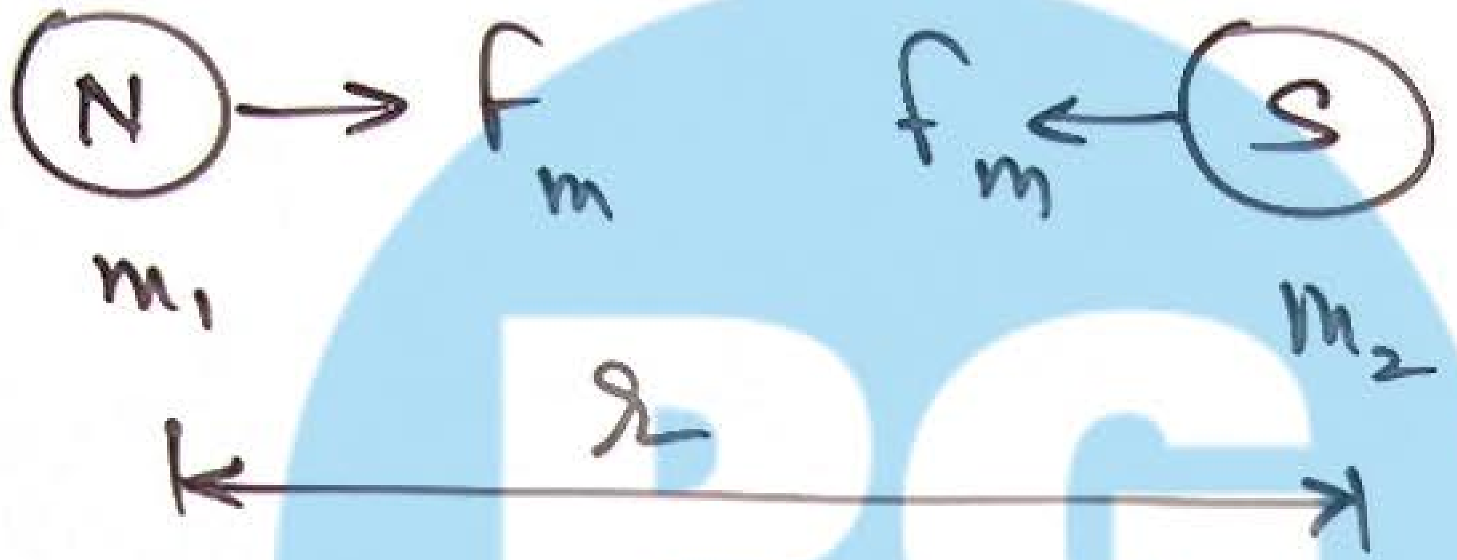
① Coulomb' Law for Magnetic Forces:

Unipole never exist in Nature

medium (μ_r)



m_1, m_2

magnetic Pole Strength

Unit - 'Amp-m'

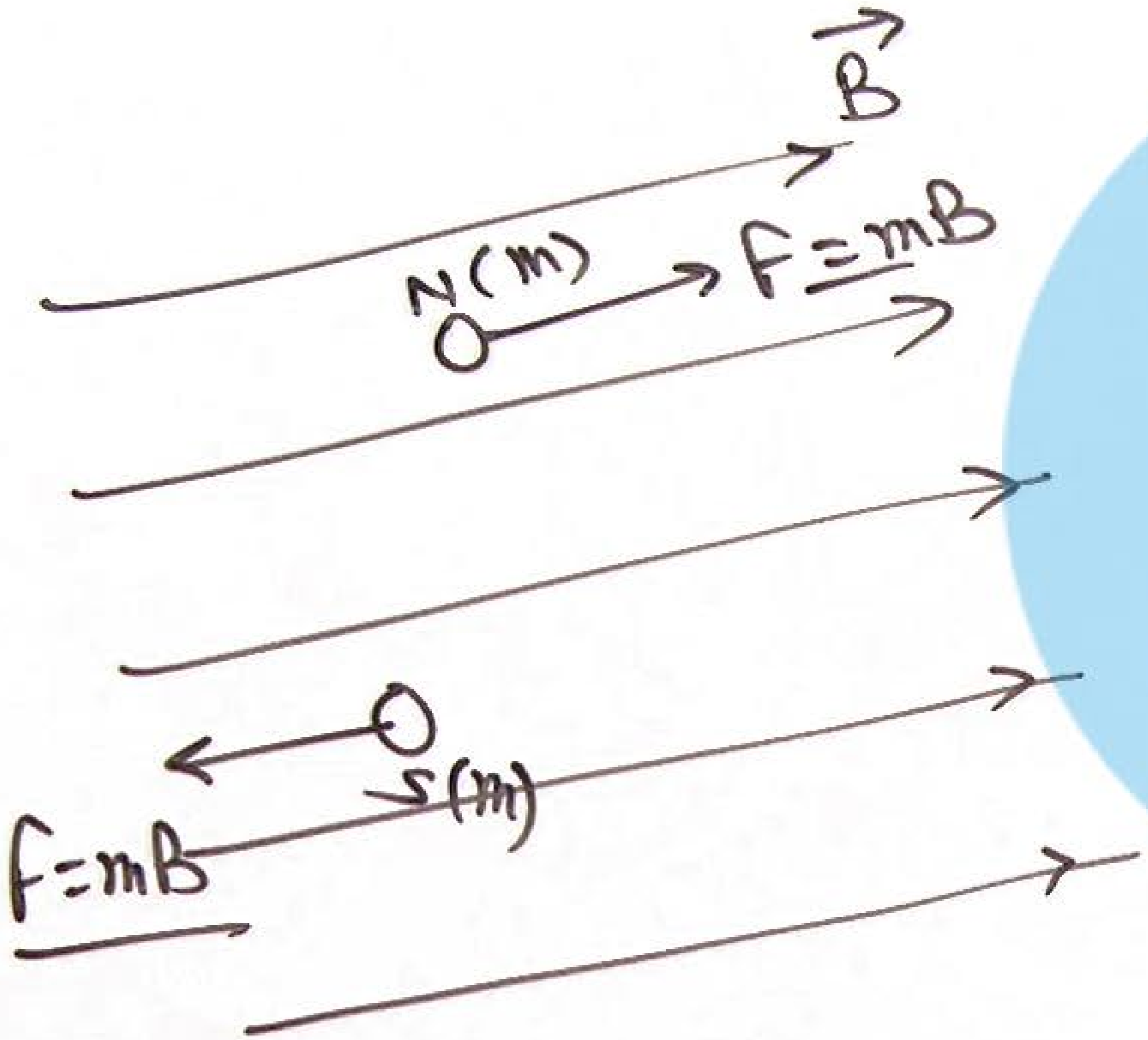
$$F_m = \frac{K m_1 m_2}{r^2} = \frac{\mu_0 \mu_r}{4\pi} \frac{m_1 m_2}{r^2}$$

$4\pi \times 10^{-7}$ SI units

$$K = \frac{\mu}{4\pi} = \frac{\mu_0 \mu_r}{4\pi}$$

CLASSICAL & TERRESTRIAL MAGNETISM

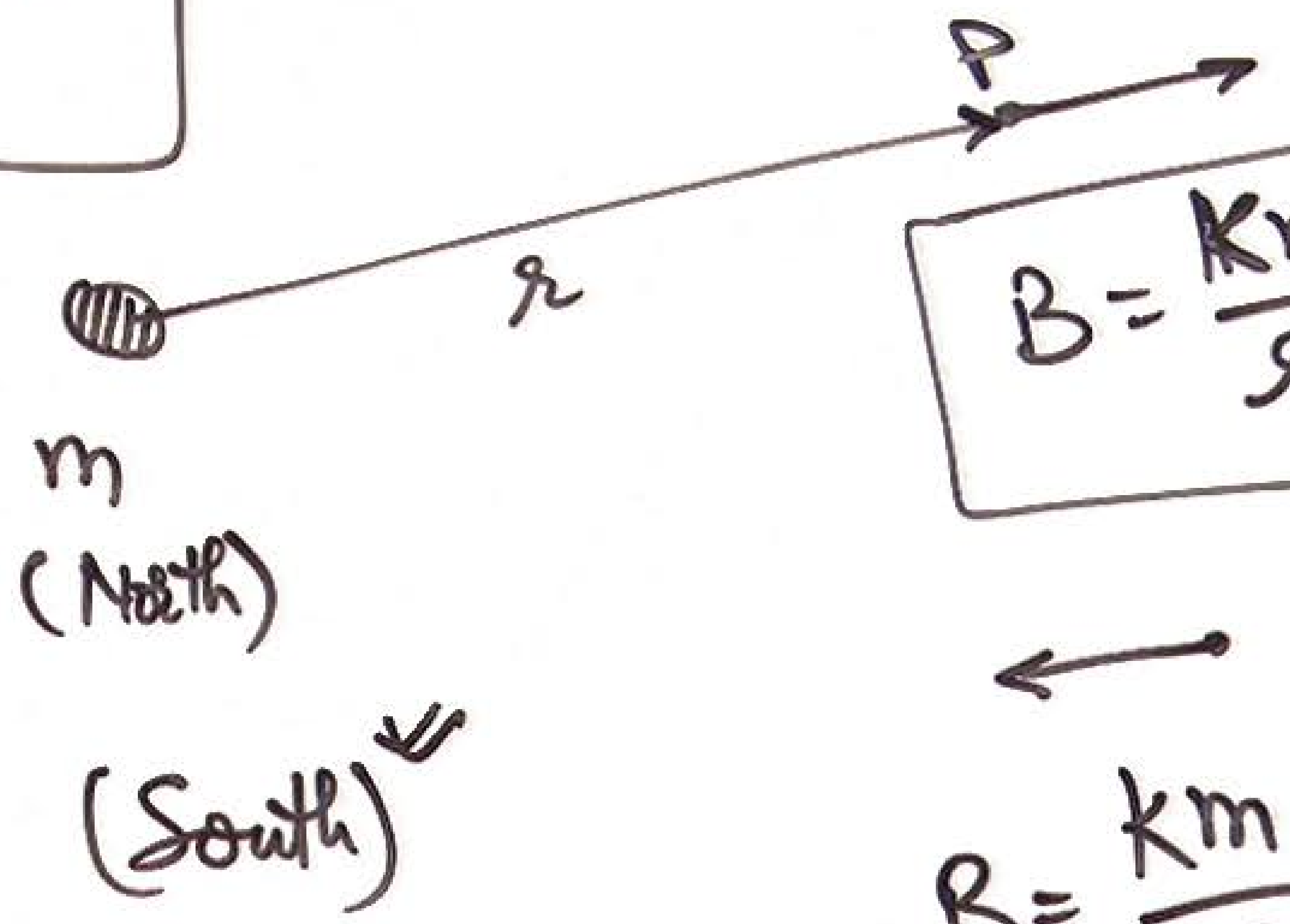
(2) Force on a magnetic pole in Magnetic Field:



(3) Magnetic Field due to a point magnetic pole:

Vectorially

$$\boxed{\vec{B} = \frac{km}{r^3}\cdot\vec{r}}$$

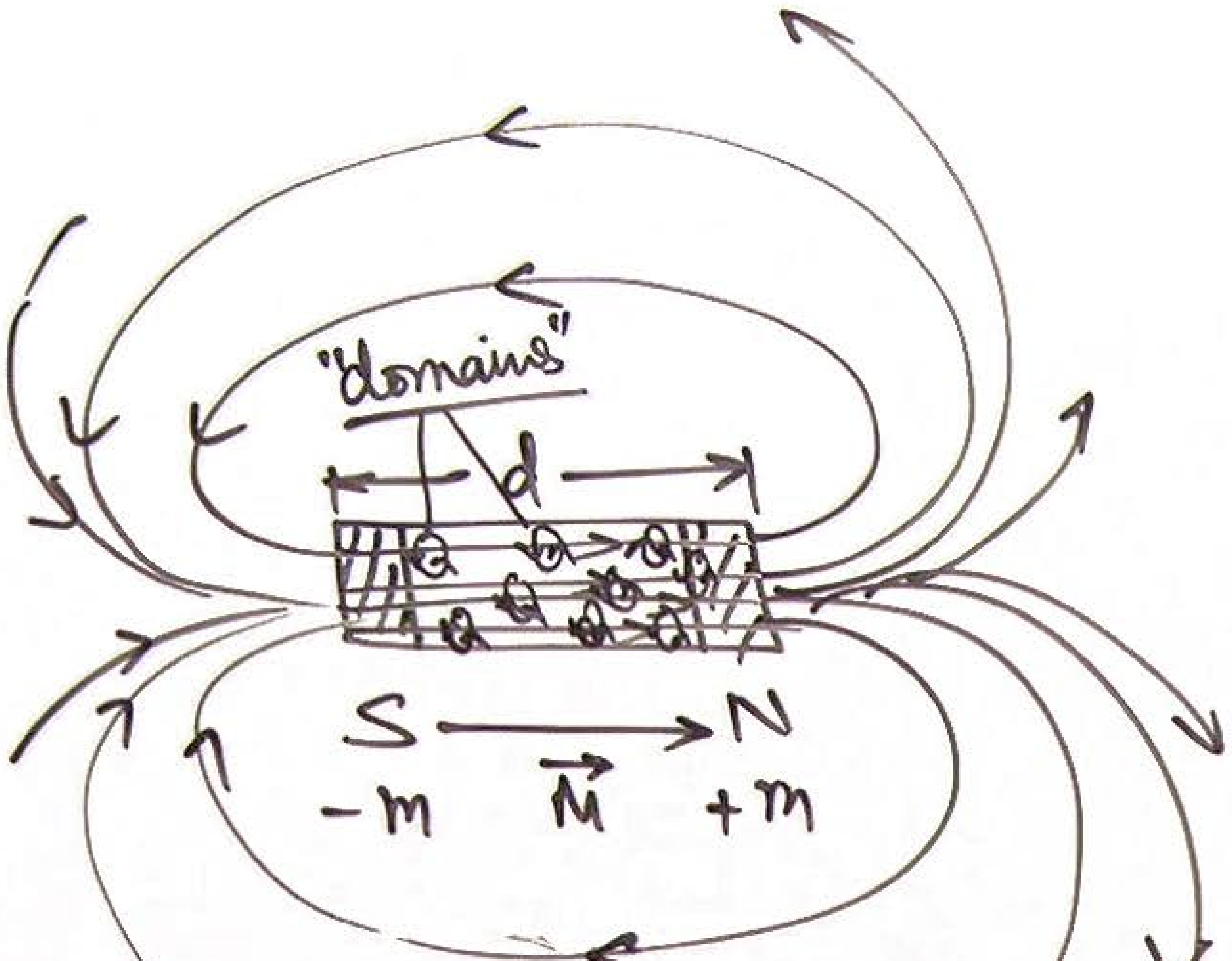


$$\boxed{B = \frac{km}{r^2}}$$

$$B = \frac{km}{r^2}$$

Classical & Terrestrial Magnetism

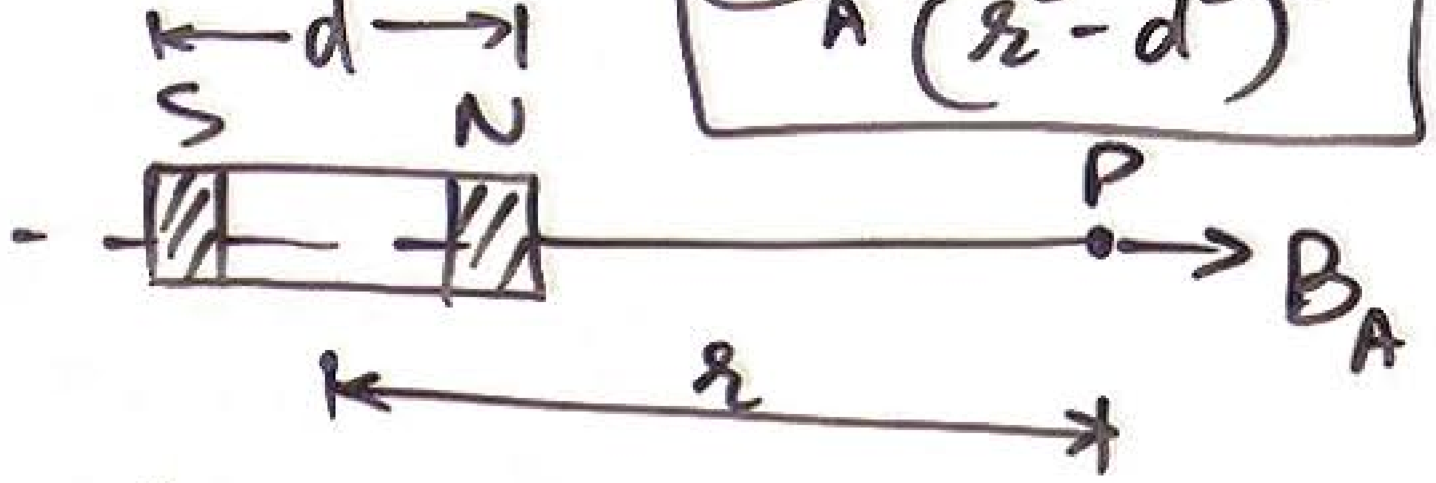
(4) Bar Magnet:



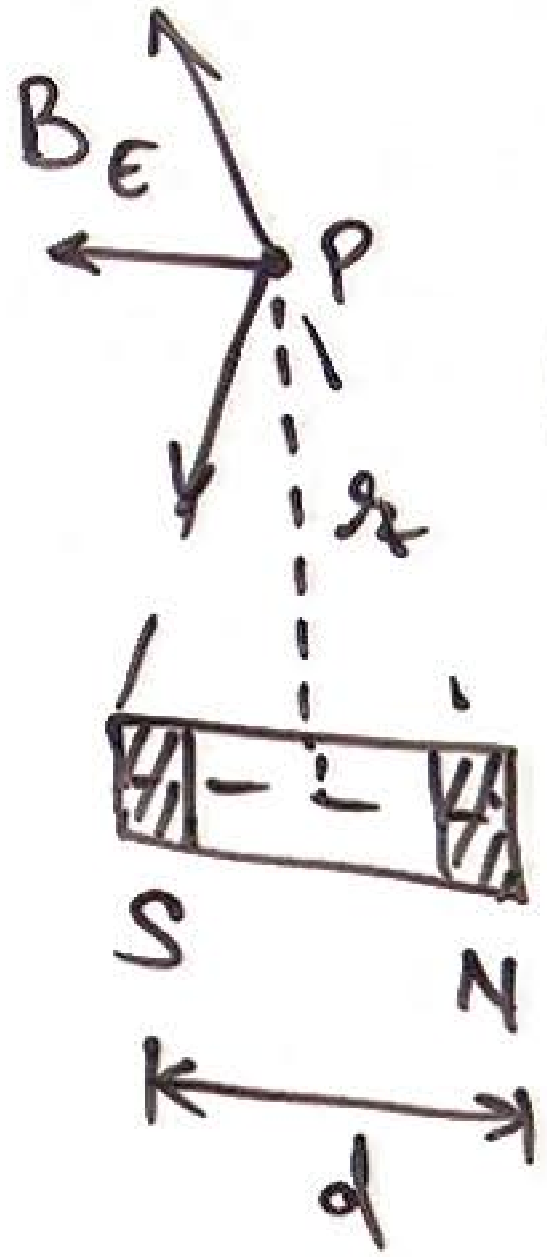
$$\boxed{\vec{M} = m\vec{d}}$$

magnetic dipole moment

(5) MF due to a Bar Magnet:

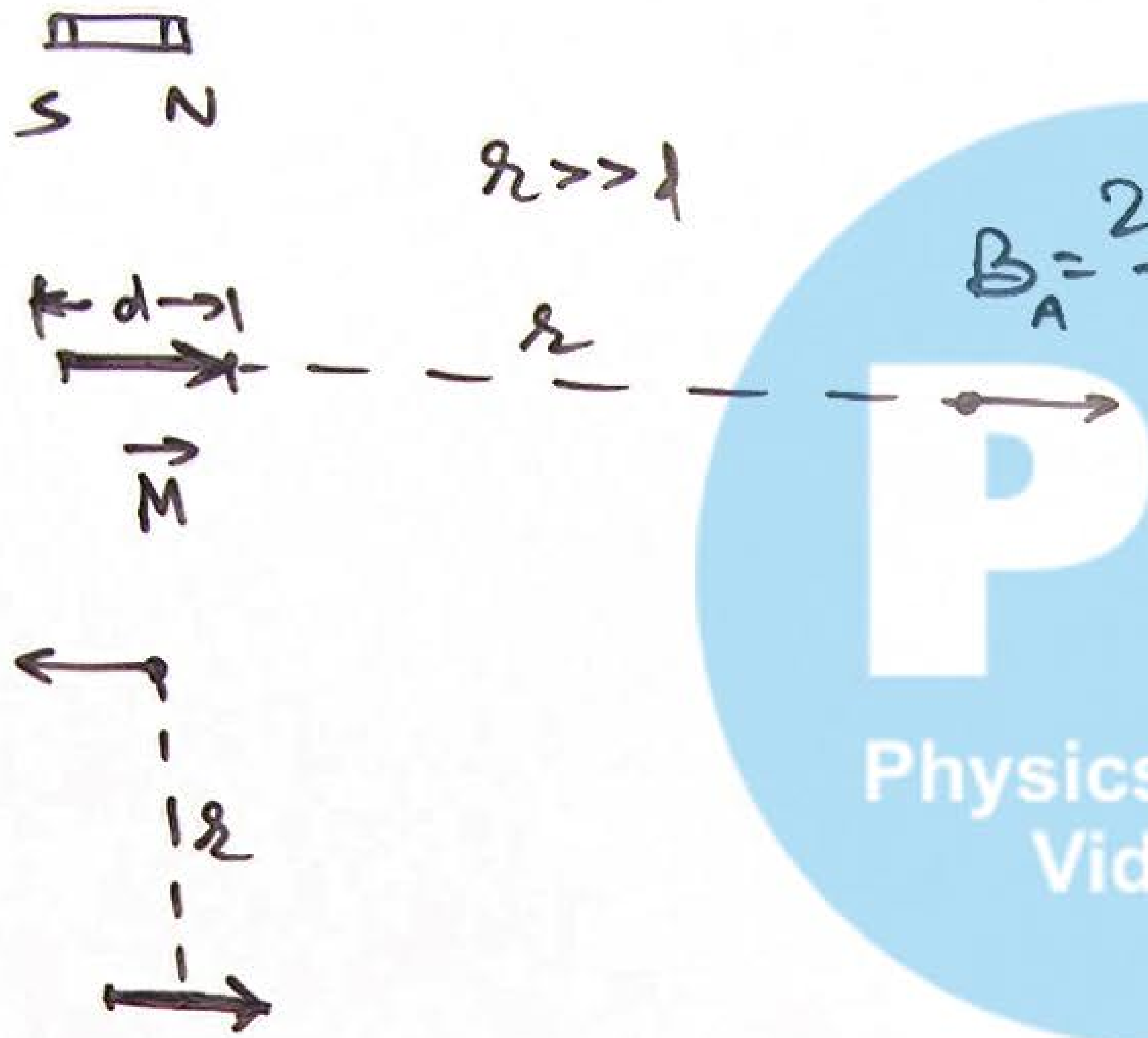


$$\boxed{B_A = \frac{2KMr}{(r^2-d^2)^2}}$$



$$\boxed{B_E = \frac{KM}{(r^2+d^2)^{3/2}}}$$

6 MF due to a Magnetic Dipole:



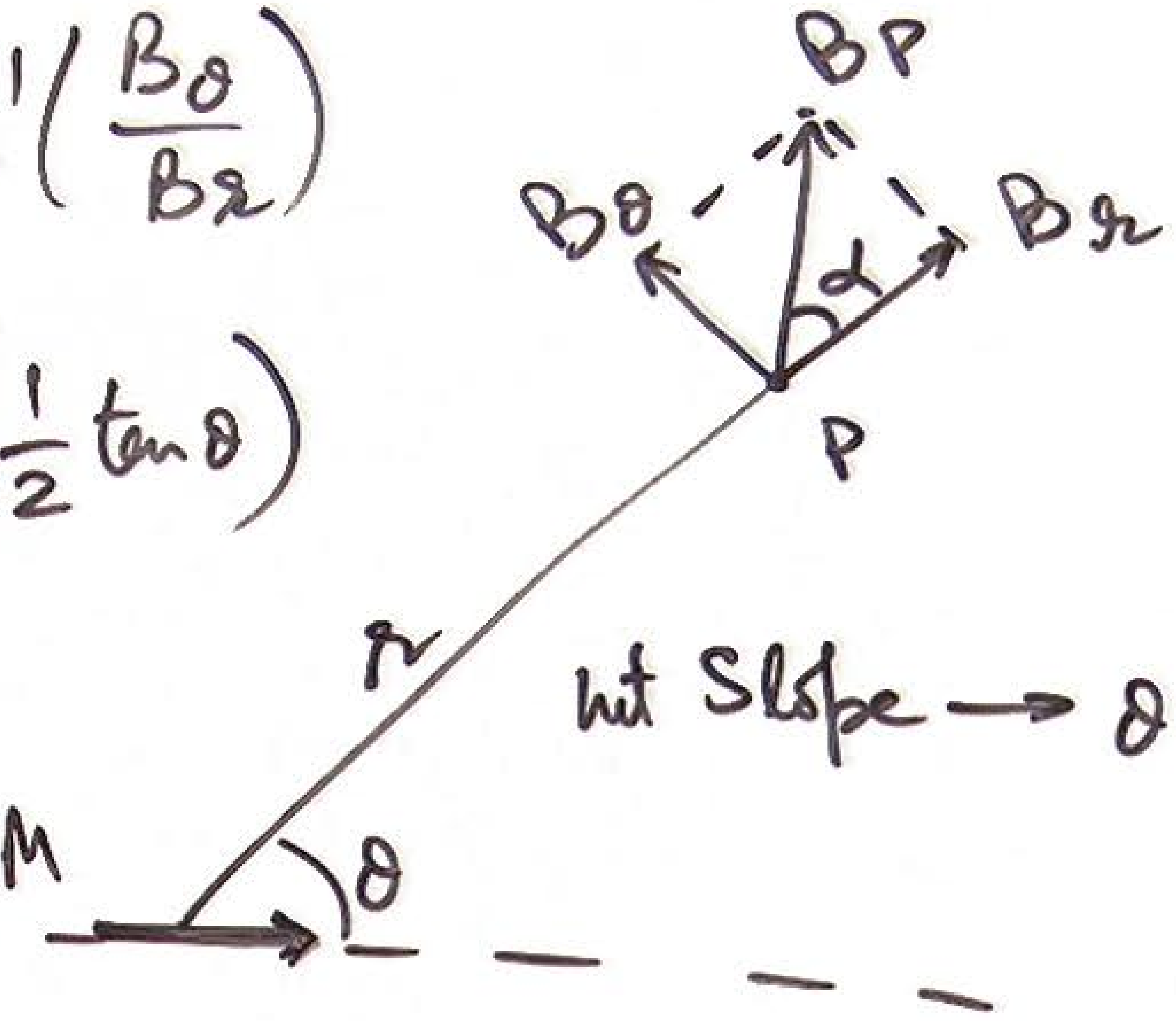
$$B_A = \frac{2KM}{r^3}$$

$$B_E = \frac{KM}{r^3}$$

$r >> d$

$$\alpha = \tan^{-1}\left(\frac{B_\theta}{B_r}\right)$$

$$\alpha = \tan^{-1}\left(\frac{1}{2}\tan\theta\right)$$

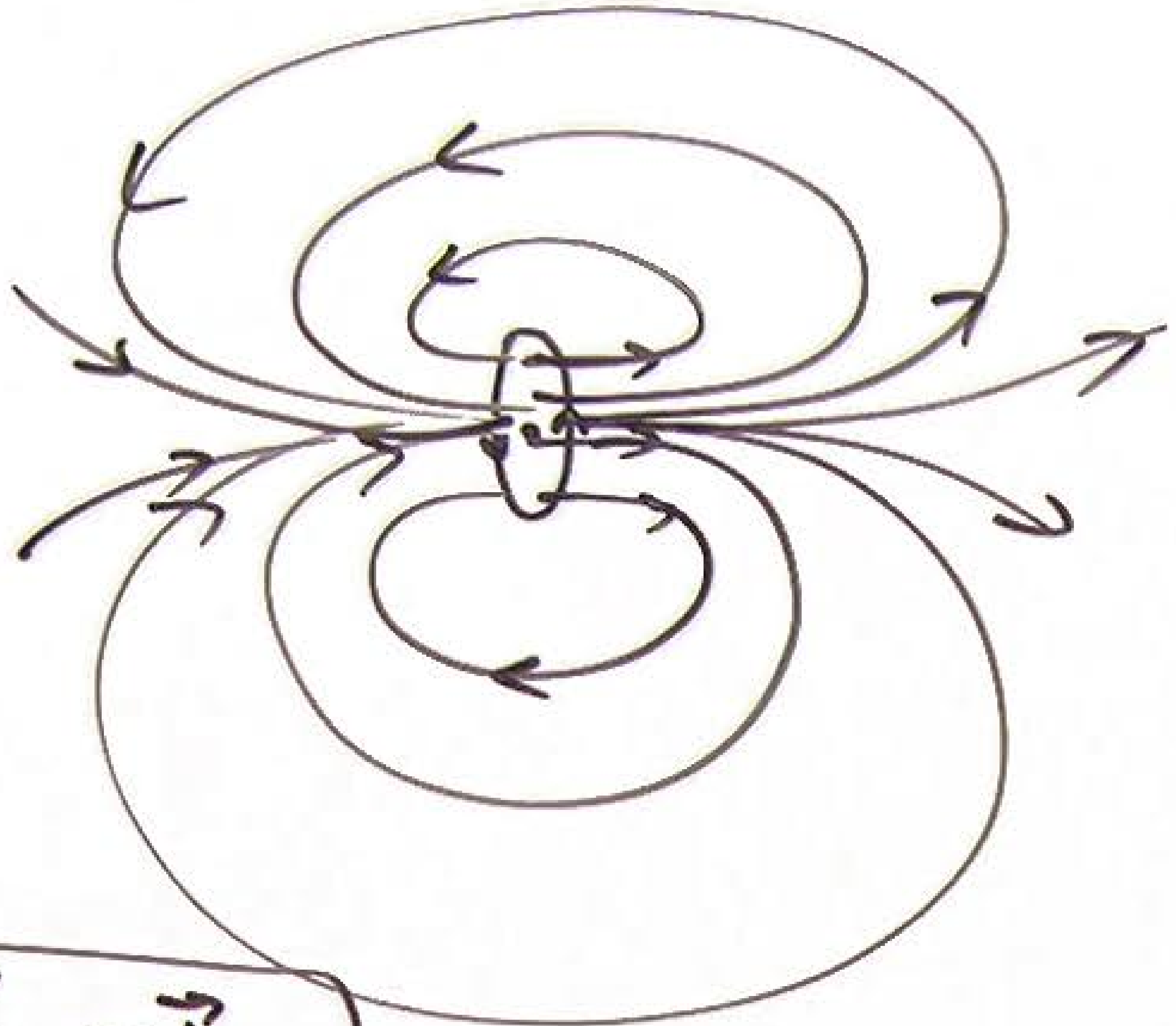


Init Slope → $\theta + \alpha$

$$B_r = \frac{2KM\cos\theta}{r^3} \; ; \; B_\theta = \frac{KM\sin\theta}{r^3}$$

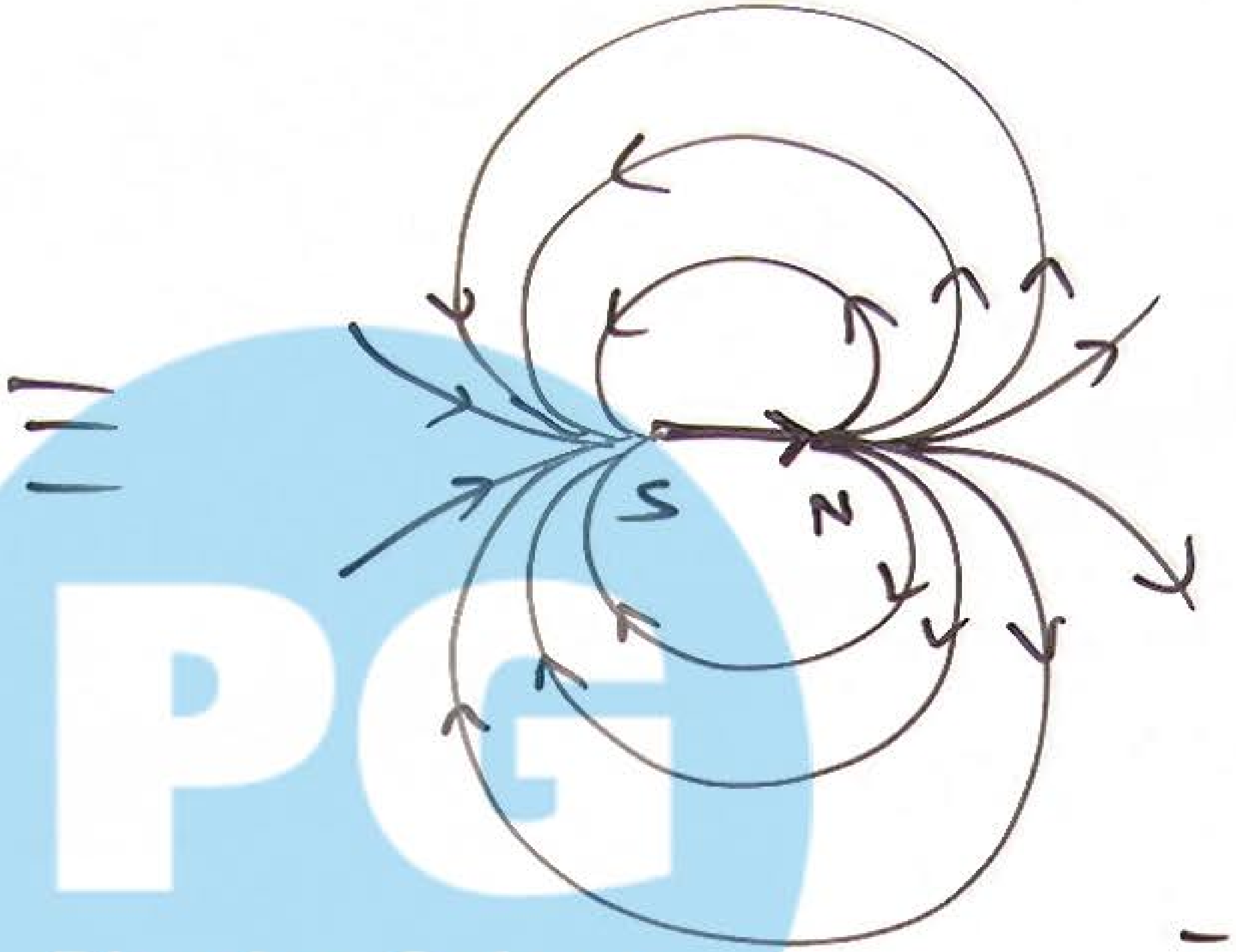
$$B_P = \sqrt{B_r^2 + B_\theta^2} = \frac{KM}{r^3}\sqrt{1+3\cos^2\theta}.$$

7 A small coil as Magnetic Dipole:



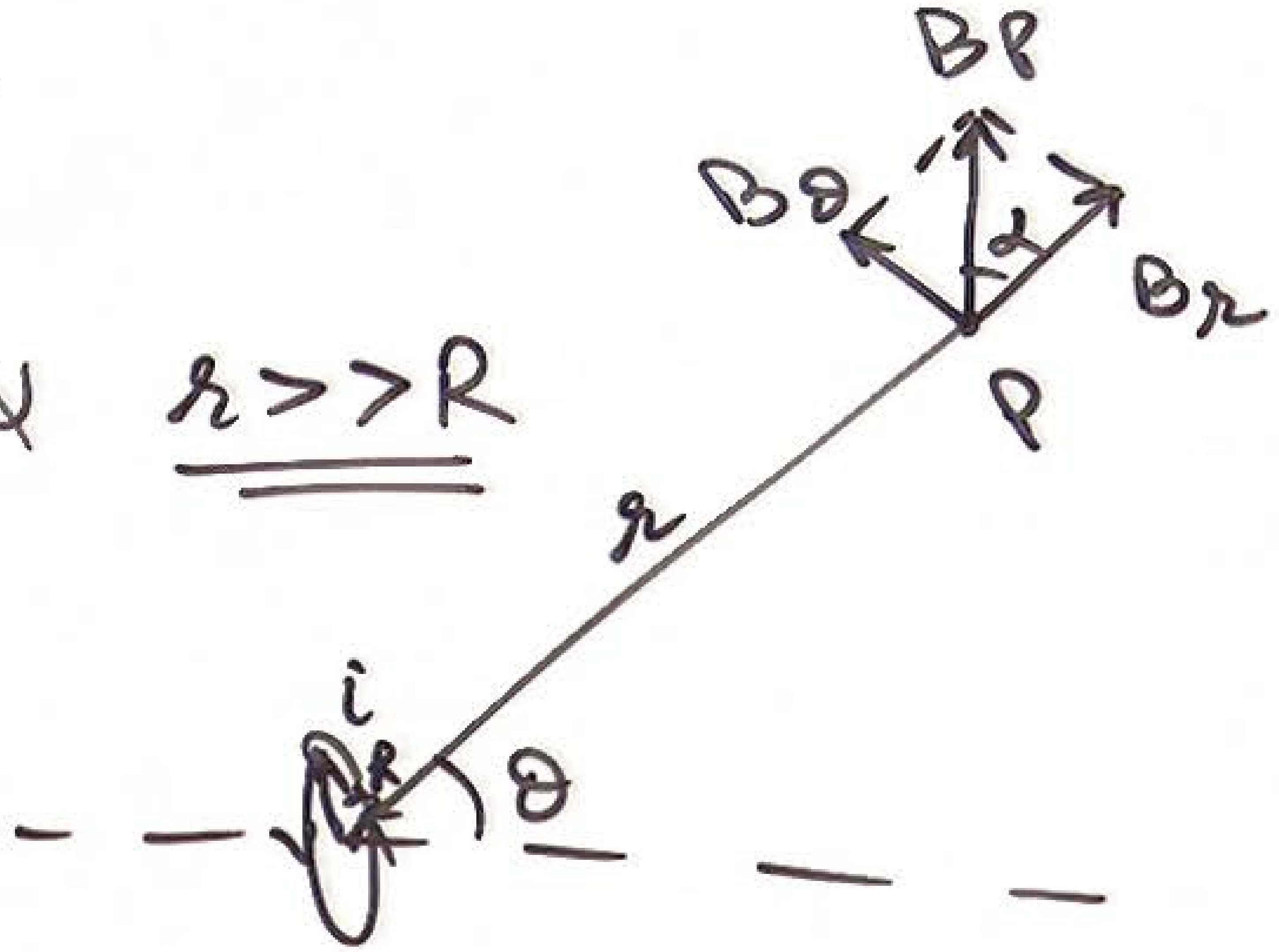
$\boxed{\vec{M} = I\vec{A}N}$

V. small sized current Carrying coil produces MF Similar to a magnetic dipole.



$\vec{M} = m\vec{d}$

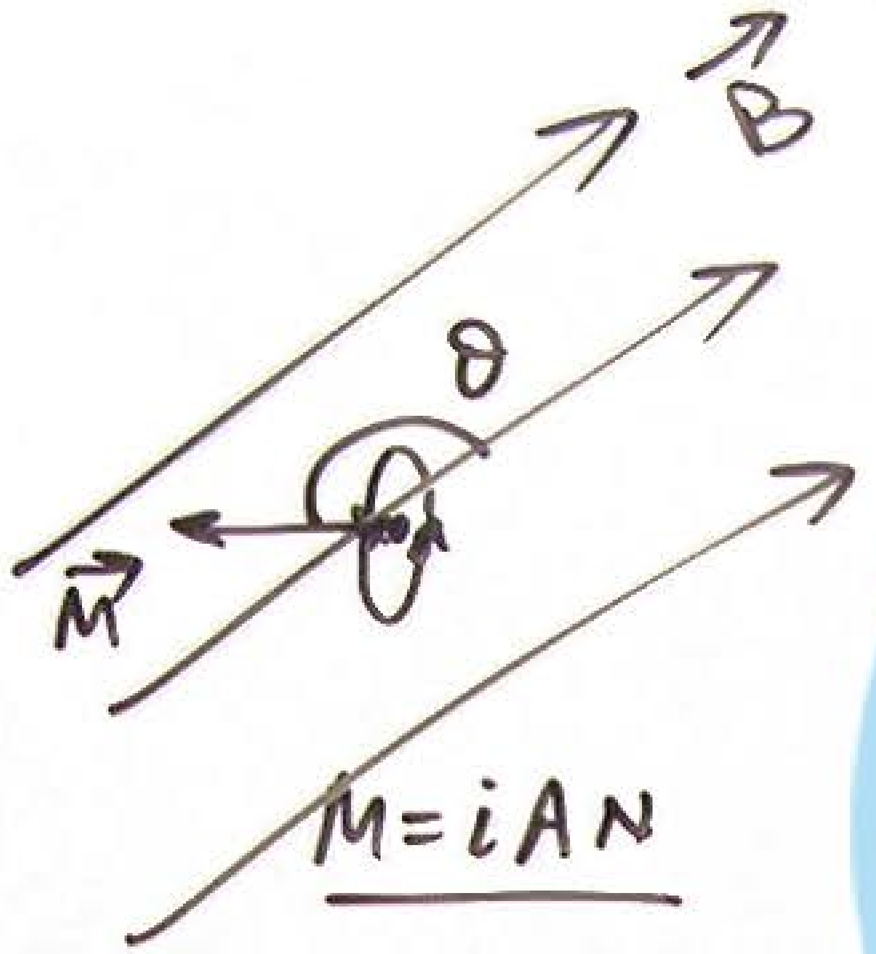
$r >> R$



$$B_r = \frac{2K(i\pi R^2)\cos\theta}{r^3}$$

$$B_\theta = \frac{K(i\pi R^2)\sin\theta}{r^3}$$

(8) Force and Torque on a small coil in MF :



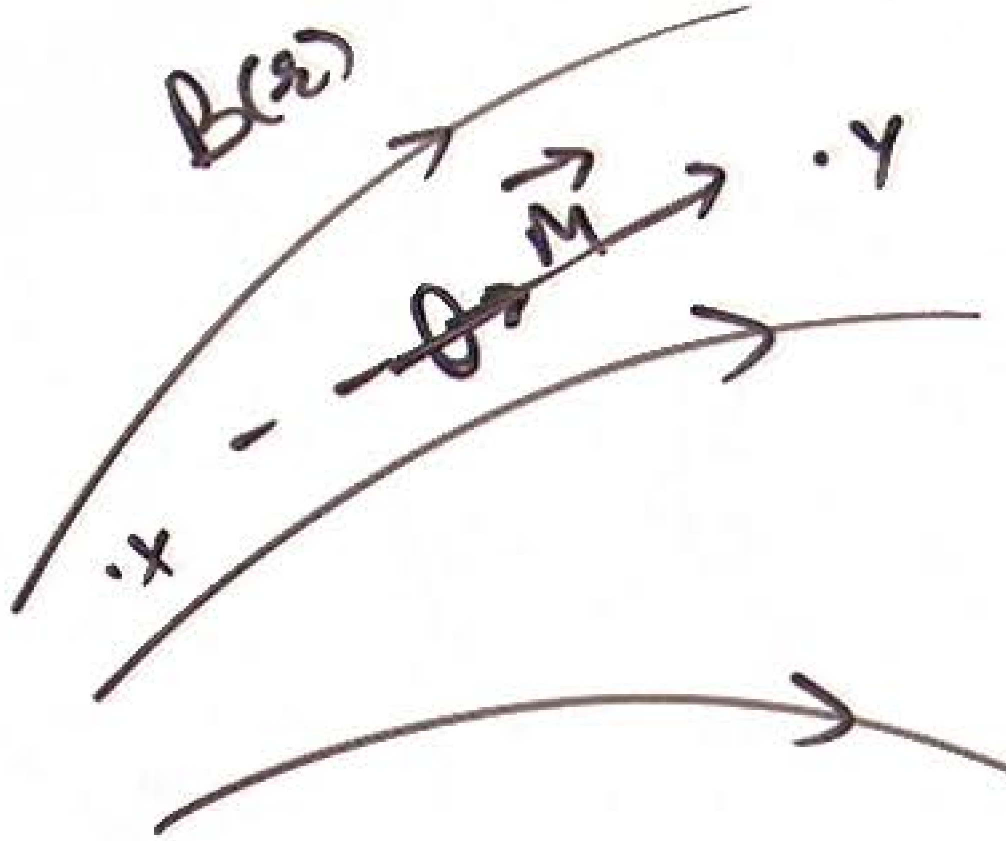
$M = iAN$

$F_{net} = 0$

$\tau_{net} = MB\sin\theta$

force on dipole $\boxed{\vec{F} = p\frac{dE}{dx}\hat{i}}$

Variable MF



force on dipole

$$\boxed{\vec{F} = M\frac{dB}{dr}\hat{r}}$$

$B_x > B_y$ along $\hat{r}$ dir.

(8A) Force between Two Magnetic Dipoles:

$$\boxed{F_D = \frac{6KM_1M_2}{r^4}}$$

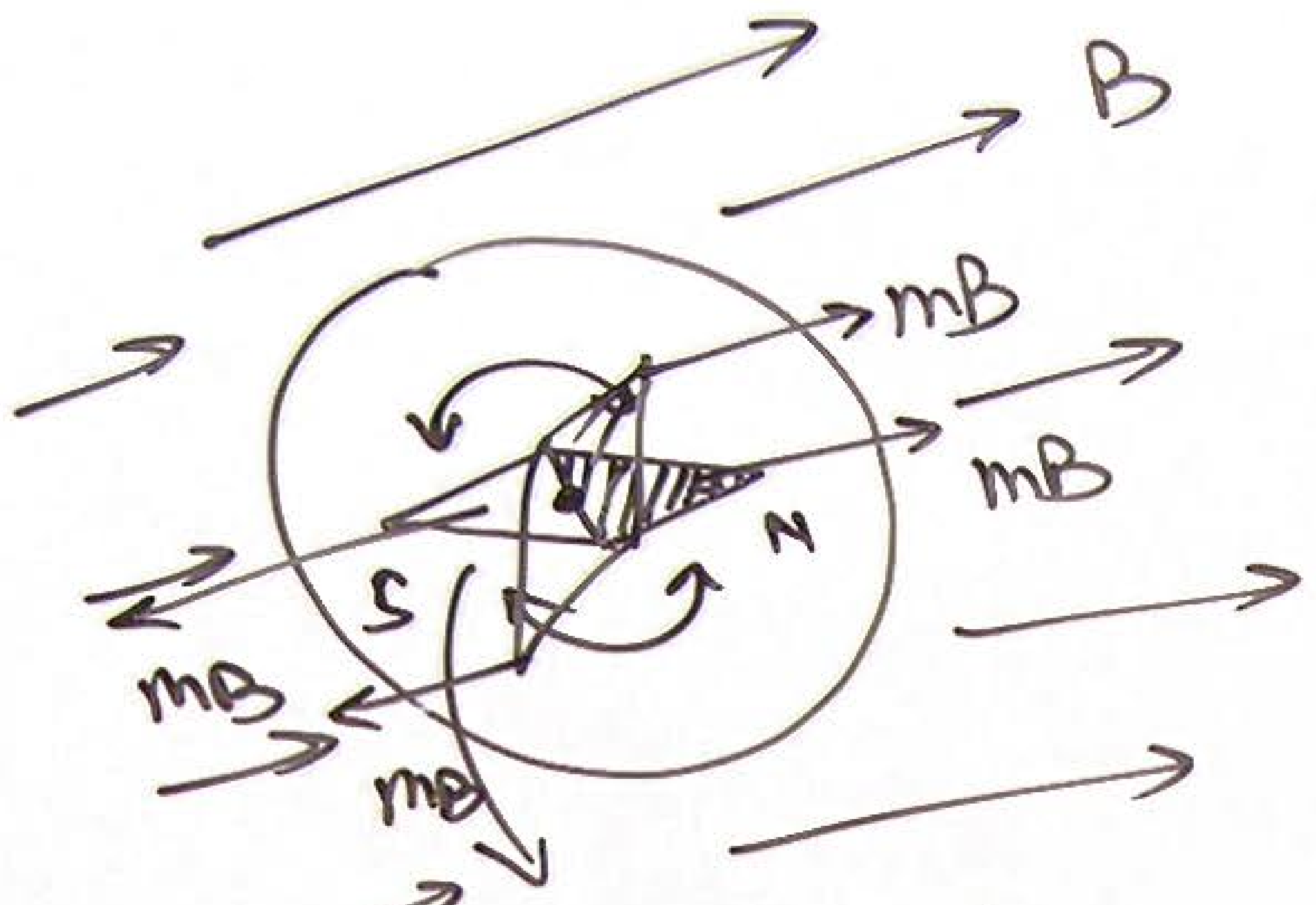
$$B_1 = \frac{2KM_1}{r^3}$$

force on M_2

$$F = M_2 \cdot \frac{dB}{dr} = M_2\left(-\frac{6KM_1}{r^4}\right)$$

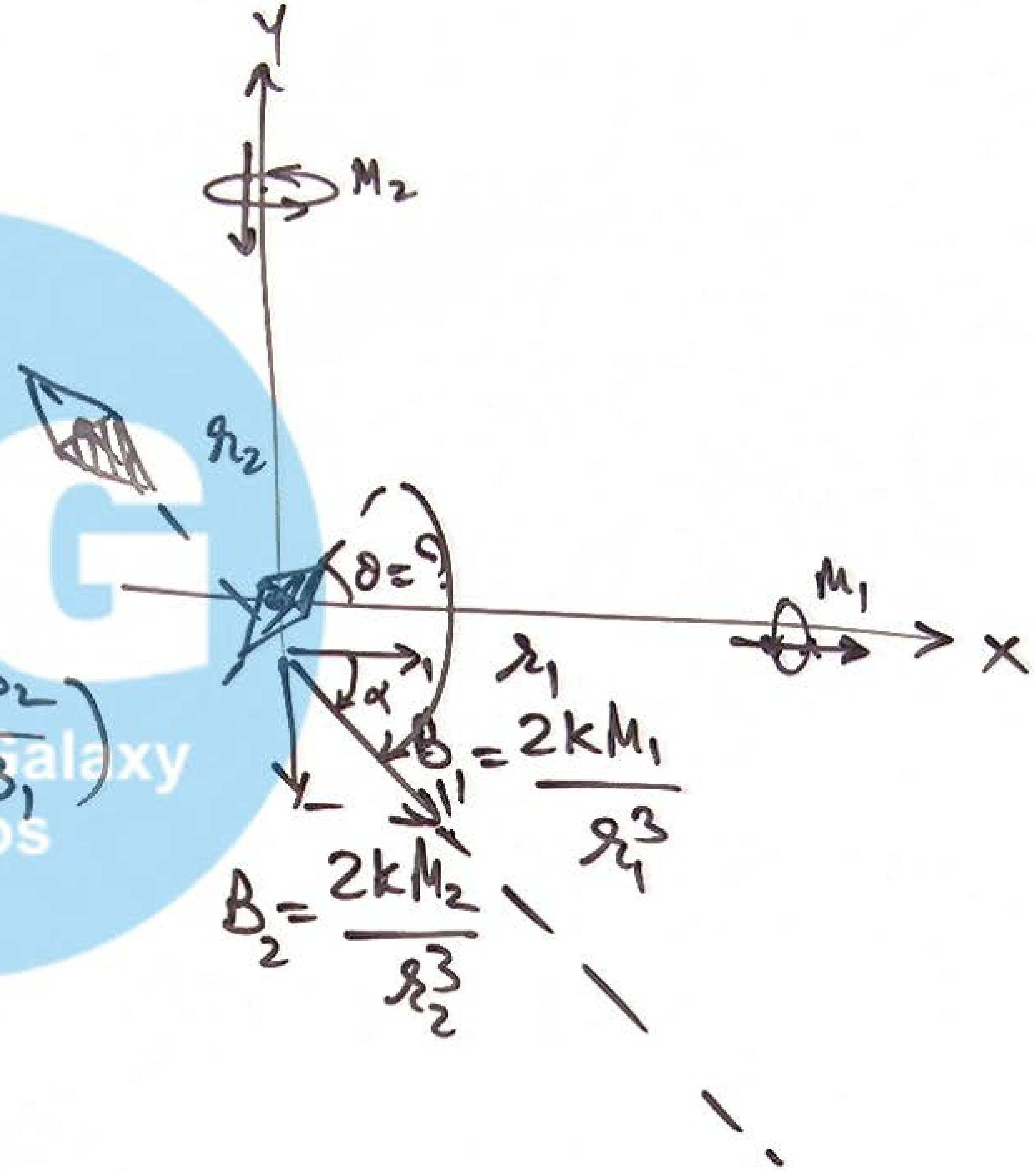
$$\Rightarrow F \propto 1/r^4$$

(9) Magnetic Needle:



Magnetic dipole
or
light small magnet

IL:



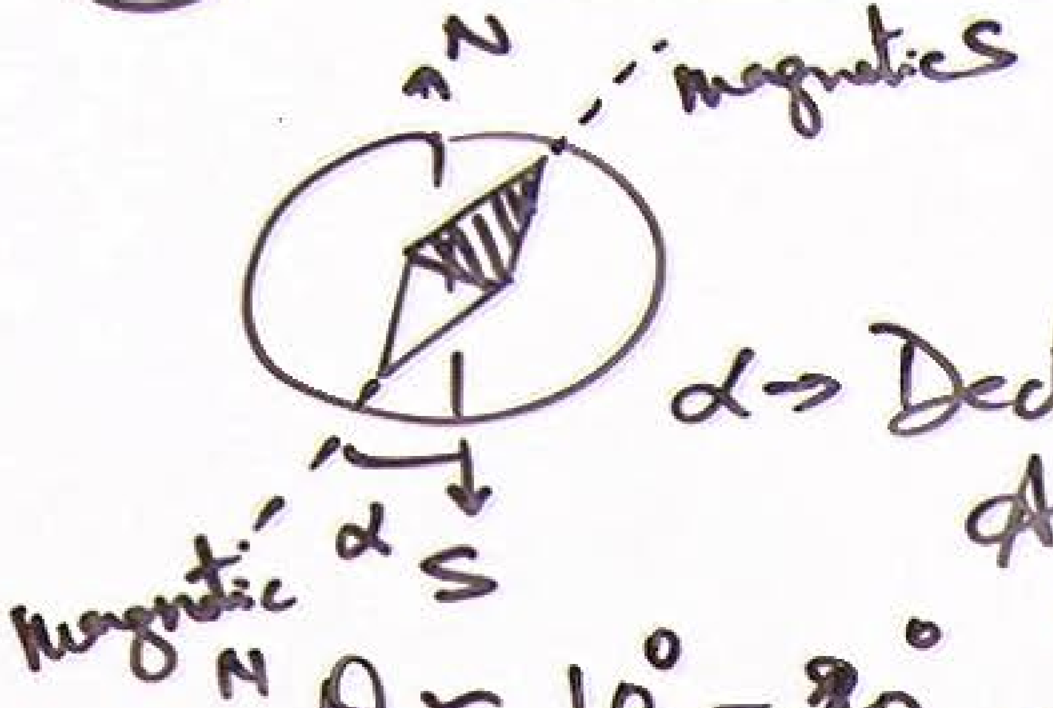
$\alpha = \tan^{-1}\left(\frac{B_2}{B_1}\right)$

Ais

$B_1 = \frac{2kM_1}{r_1^3}$

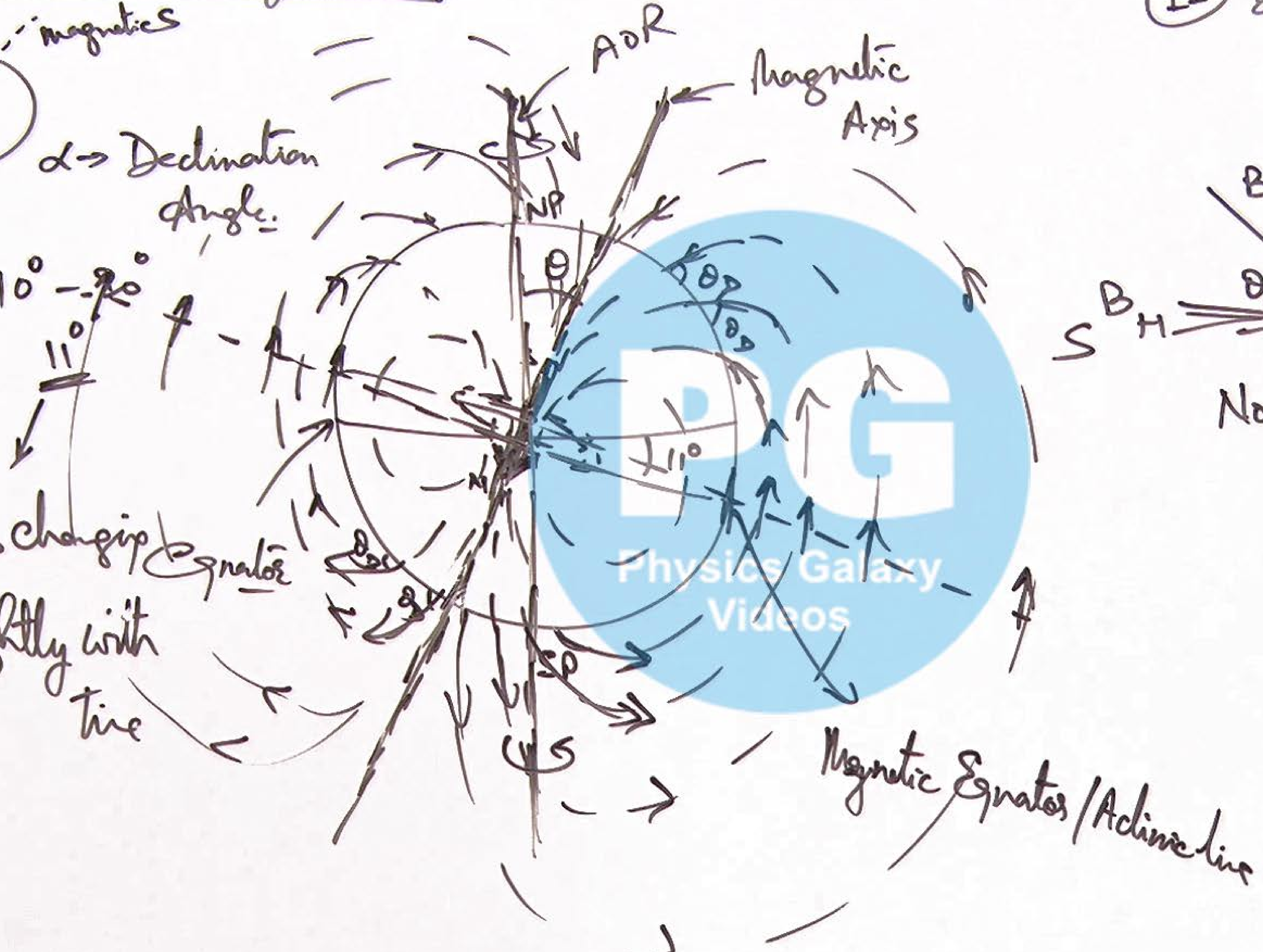
$B_2 = \frac{2kM_2}{r_2^3}$

(10) Terrestrial Magnetism:



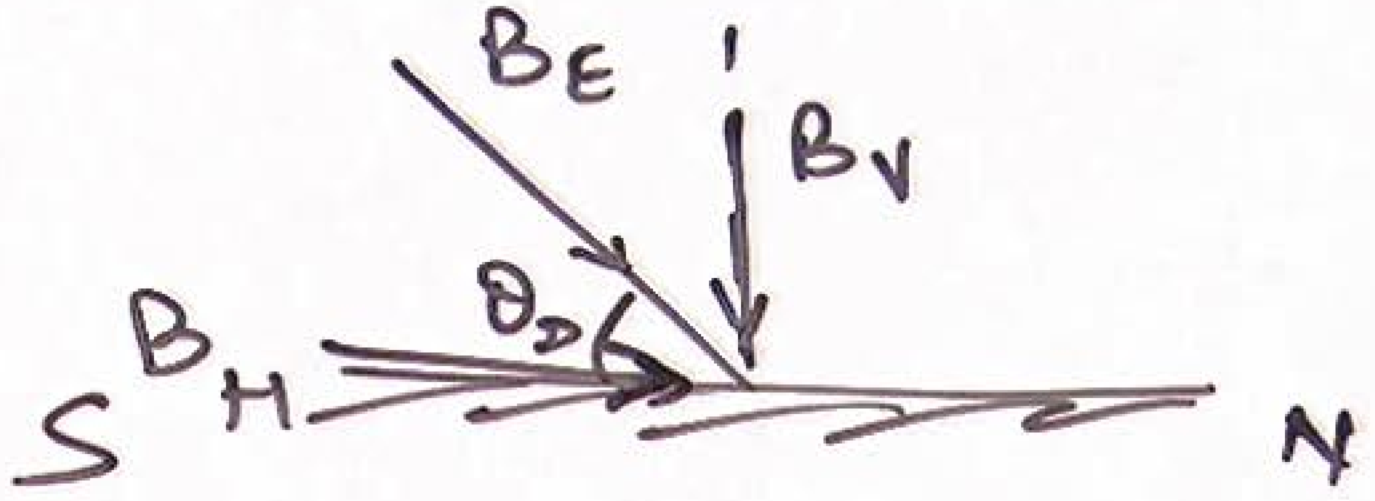
$\alpha \rightarrow$ Declination Angle.

$\theta \simeq 10° - 20°$



Keeps changing slightly with time

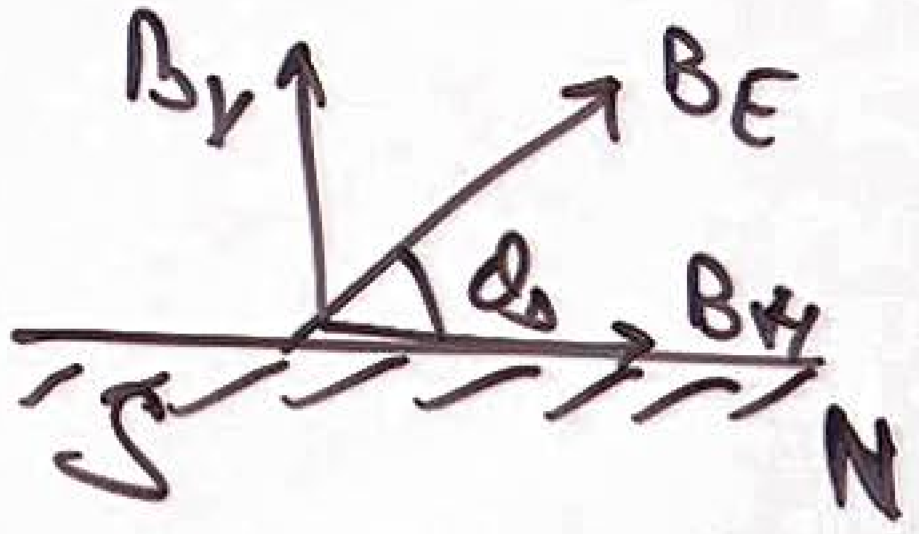
(11) Dip Angle.



Northern Hemisphere

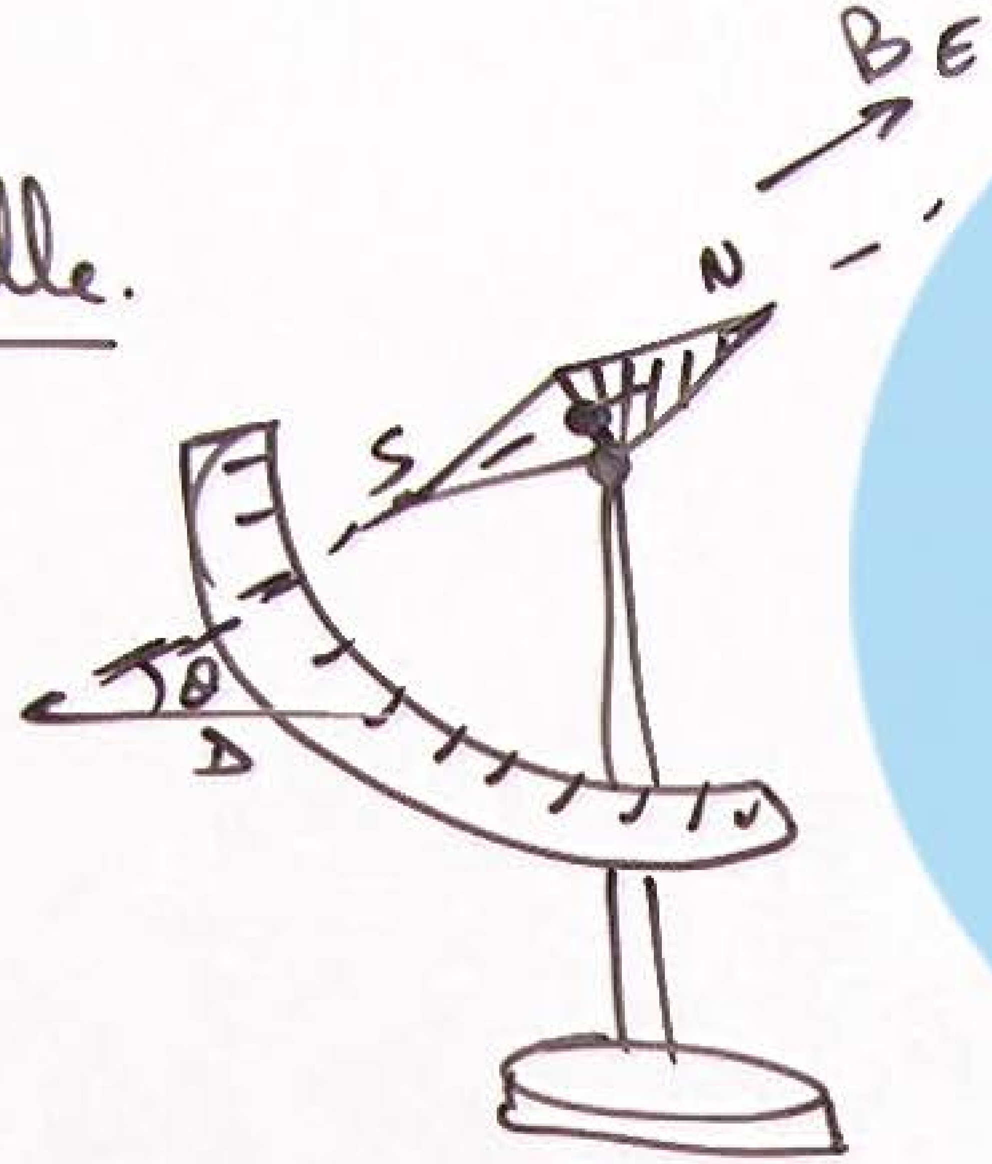
$B_V = B_E \sin\theta$

$B_H = B_E \cos\theta$



(12) Apparent Dip angle at a point on Earth Surface :

Dip needle.



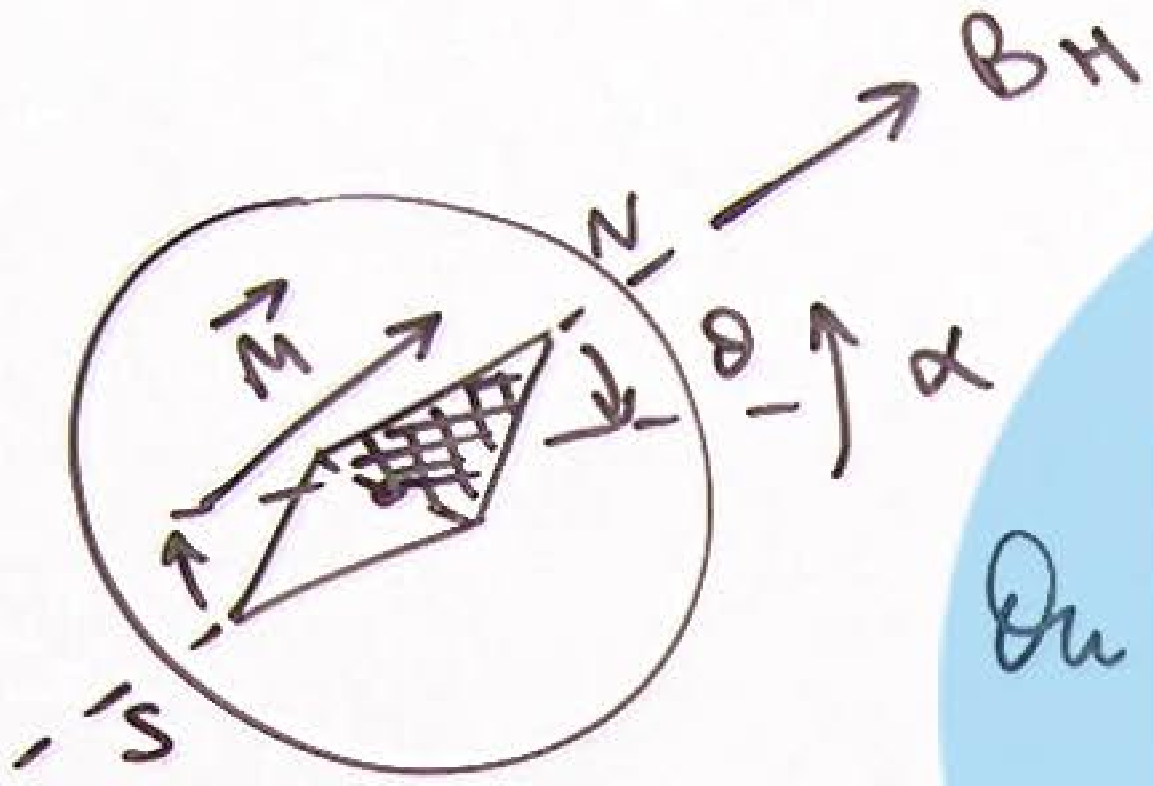
If a Dip needle is placed in plane at an angle α to Magnetic meridian & its reading is θ_{AD}.

At pt of Dip needle

$$B_H' = B_H \cos\alpha$$

$$\Rightarrow \tan\theta_{AD} = \frac{B_V}{B_H \cos\alpha} \Rightarrow \boxed{\theta_D = \tan^{-1}\left[\cos\alpha \cdot \tan\theta_{AD}\right]}$$

(13) Oscillations of a Compass Needle:



Horizontal plane.

On slight tilt of needle, restoring torque on it is

$$\tau_R = MB_H \sin\theta$$

Ang acc^n $\alpha = \frac{\tau_R}{I} = -\frac{MB_H}{I}\theta$ [Small angle $\Rightarrow \sin\theta \simeq \theta$]

Compare with $\alpha = -\omega^2\theta$

$$\omega = \sqrt{\frac{MB_H}{I}}$$

$$T = 2\pi\sqrt{\frac{I}{MB_H}} = 2\pi\sqrt{\frac{I}{MB_E\cos\theta_D}}$$

14 Magnetic Field Vectors $\vec{B}$, $\vec{H}$ and $\vec{I}$:

MF Induction
mag field
(Tesla) or (Wb/m²)

"Magnetizing field."
Magnetic intensity
(Amp/m)

for material

Magnetization
or
Intensity of Magnetization
(Amp/m)

net MF.

$\vec{B} = \mu_0 \mu_r \vec{H}$

$\vec{B} = \underbrace{\mu_0 \vec{H}}_{\text{ext field}} + \underbrace{\mu_0 \vec{I}}_{\text{mat field.}} \rightarrow \cancel{\mu_0} \mu_r \vec{H} = \cancel{\mu_0} \vec{H} + \cancel{\mu_0} \vec{I}$

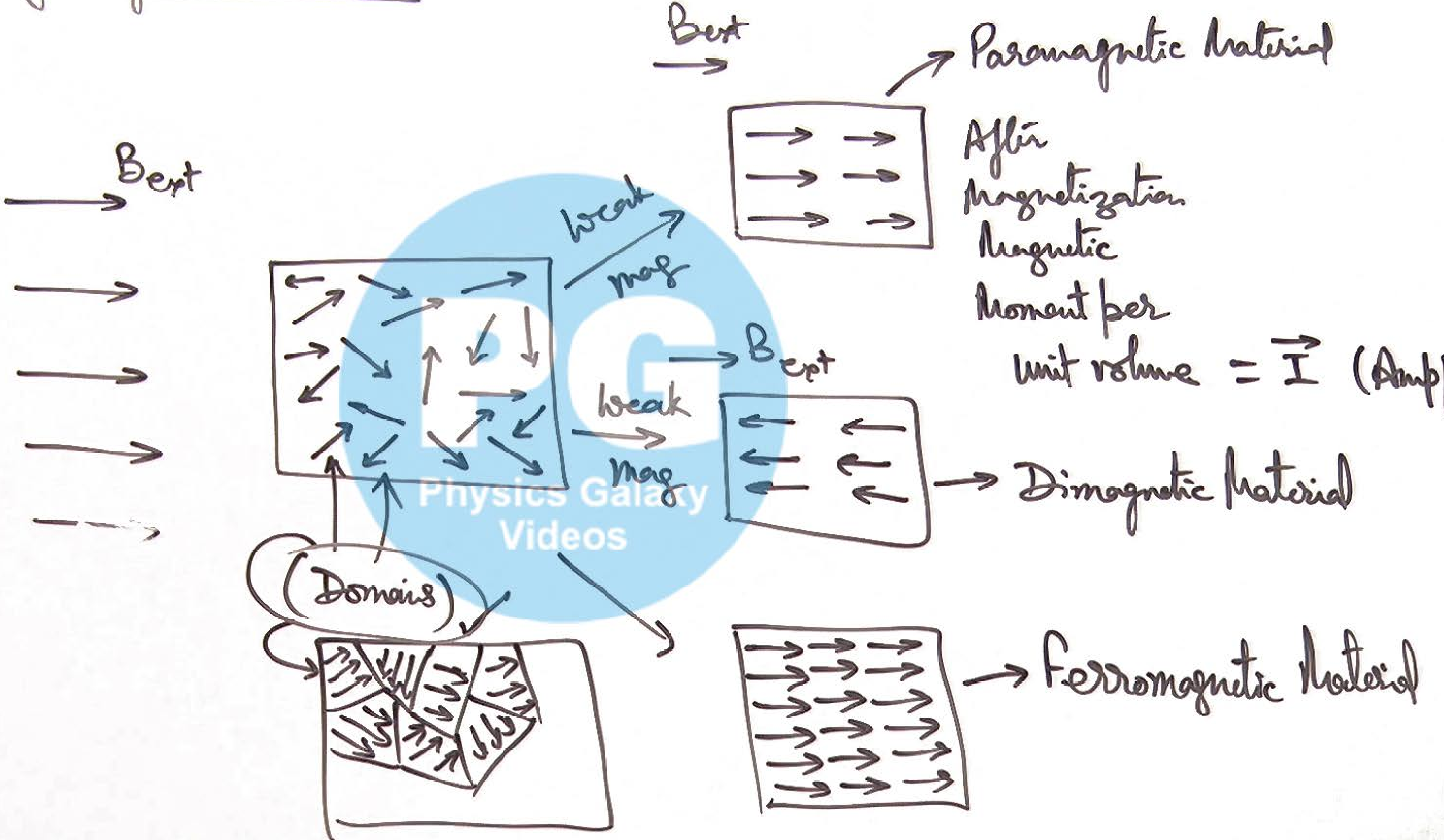
$\boxed{\mu_r = 1 + \chi}$

for a material

Magnetic Susceptibility

$\chi = \frac{I}{H}$

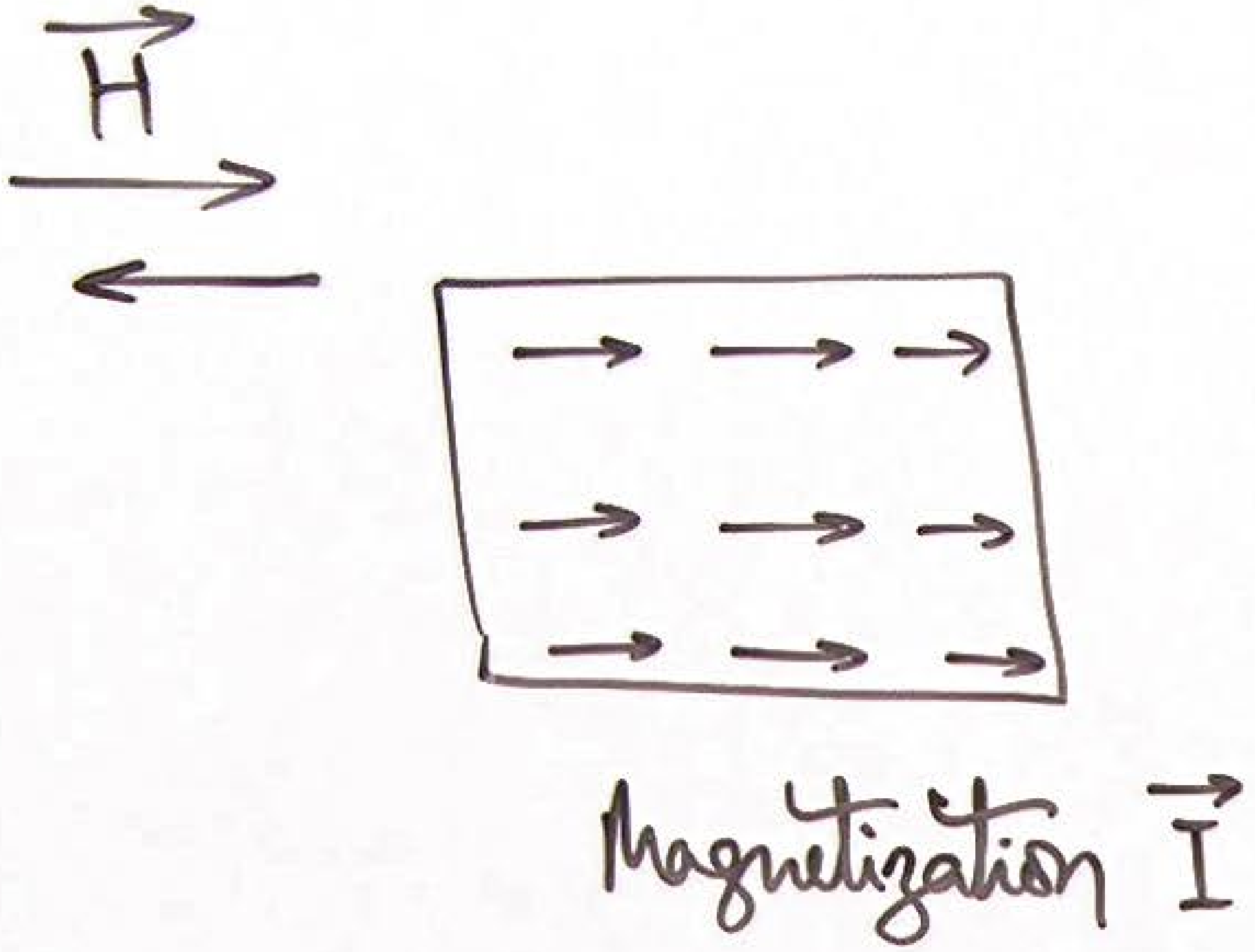
(15) Properties of Magnetic Materials :



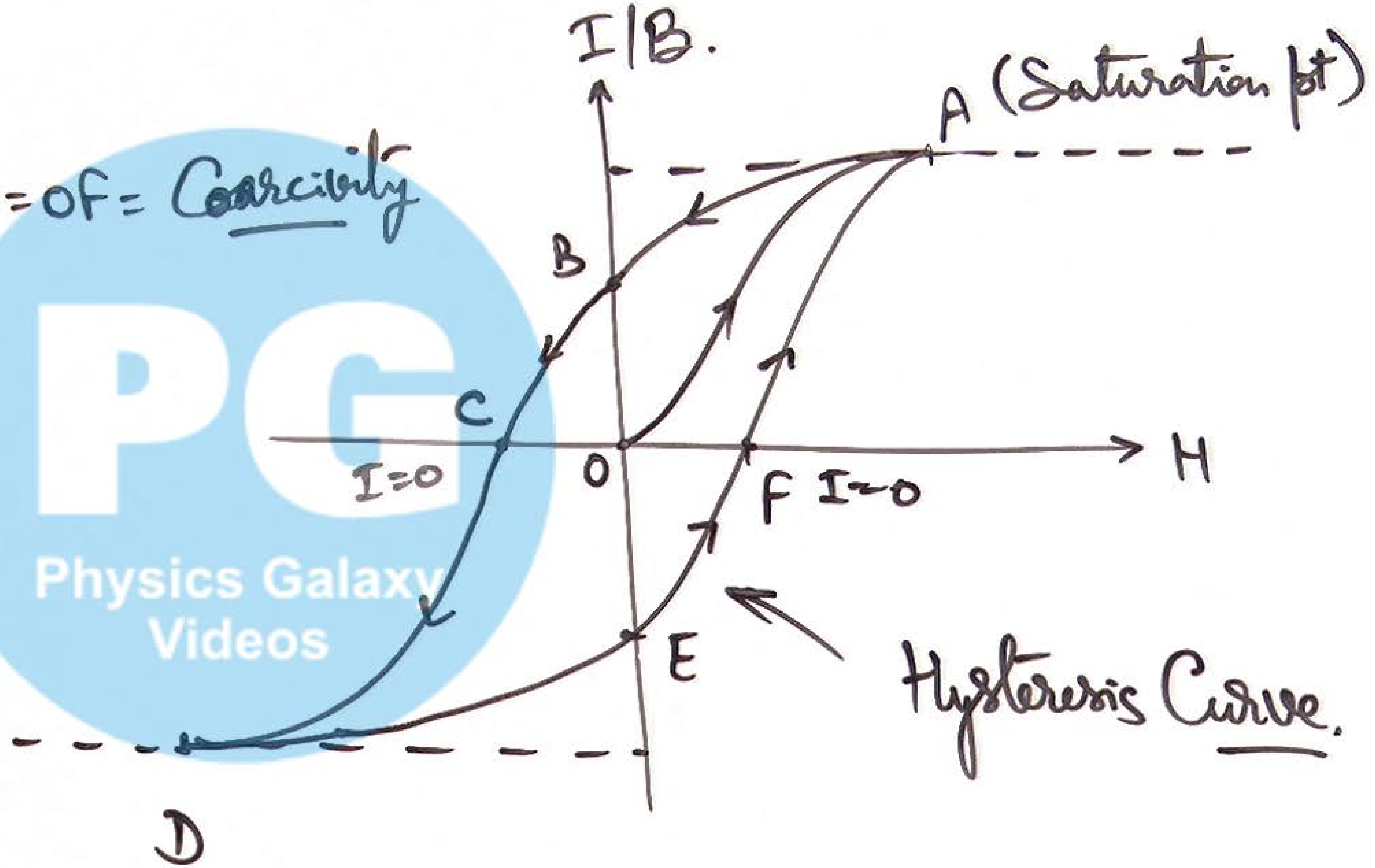
Align
Magnetization
Magnetic Moment per unit volume $= \vec{I}$ (Amp/m)

16 Magnetic Hysteresis:

OE = OB → Retentivity



OC = OF = Coercivity



Hysteresis Curve.